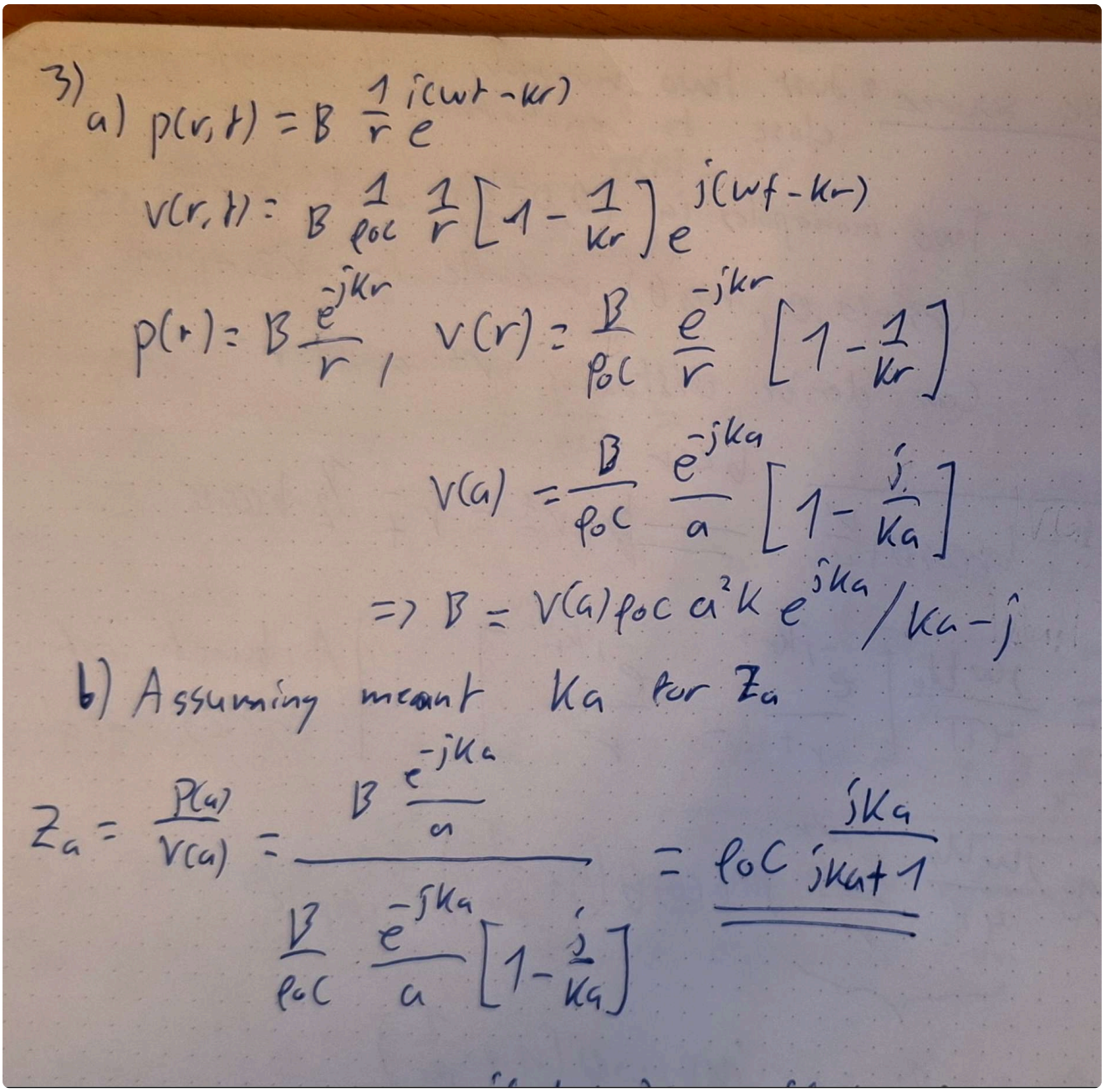
Problem 3)

a and part of b)



3) a) $p(r,t) = B \frac{1}{r} e^{j(\omega t - kr)}$

$v(r,t) = B \frac{1}{\rho_0 c} \frac{1}{r}\left[1 - \frac{1}{jkr}\right] e^{j(\omega t - kr)}$

$p(r) = B\frac{e^{-jkr}}{r}$, $v(r) = \frac{B}{\rho_0 c}\frac{e^{-jkr}}{r}\left[1 - \frac{1}{jkr}\right]$

$v(a) = \frac{B}{\rho_0 c}\frac{e^{-jka}}{a}\left[1 - \frac{j}{ka}\right]$

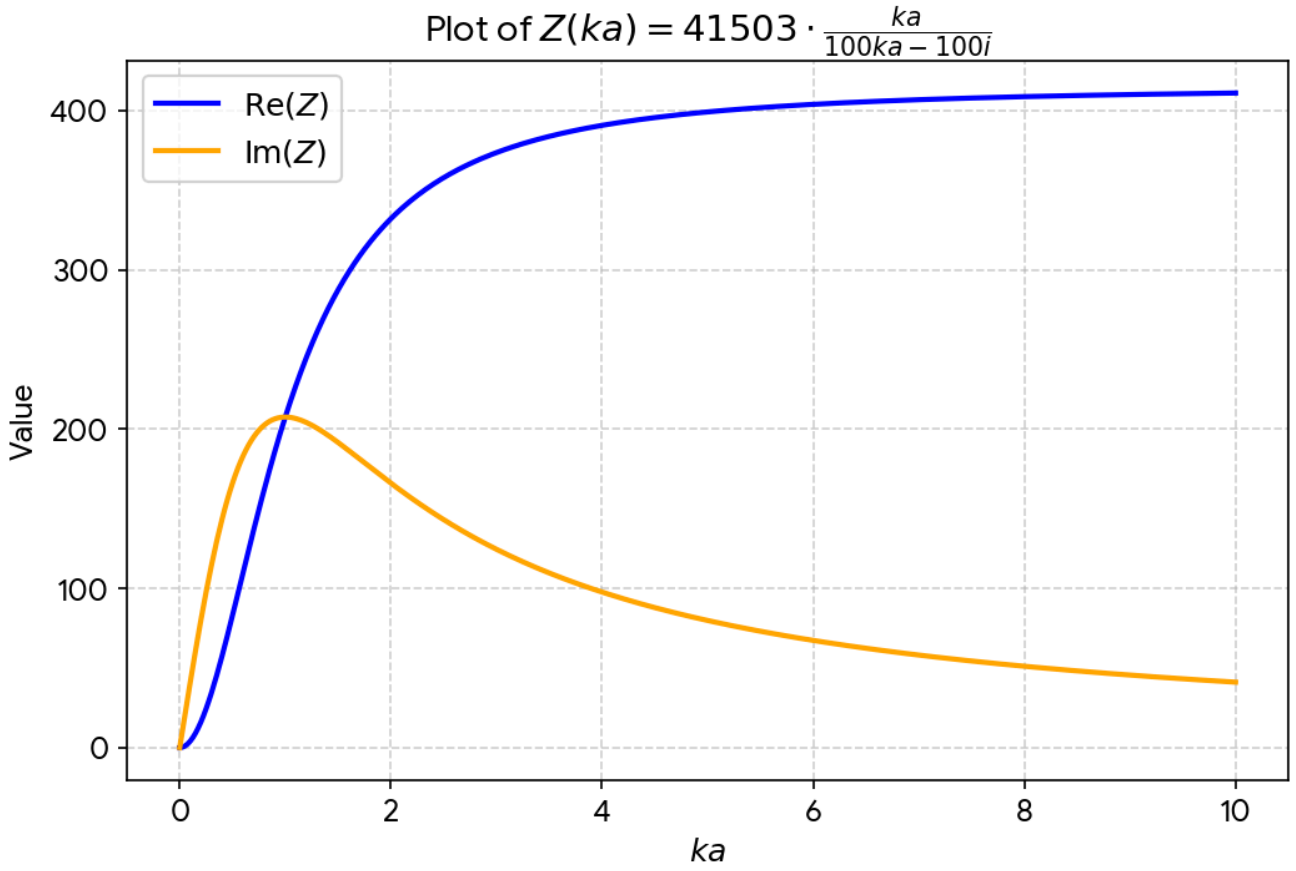
$\Rightarrow B = v(a)\rho_0 c\, a^2 k\, e^{jka} / ka - j$

b) Assuming meant ka for Z_a

$$Z_a = \frac{p(a)}{v(a)} = \frac{B\frac{e^{-jka}}{a}}{\frac{B}{\rho_0 c}\frac{e^{-jka}}{a}\left[1 - \frac{j}{ka}\right]} = \rho_0 c \frac{jka}{jka + 1}$$

b)

Plot of $Z_a(ka)$



The imaginary part reaches it peak at $ka = 1$, both start at 0 for $ka = 0$ and the imaginary part goes toward 0 as $ka \to \infty$ and the real part goes toward 415.

This means that as the frequencies get higher and higher, the sound waves reach a stable amount of attenuation, but also become less phase-shifted and stores less energy. Similar to a lowpass filter in the real part.

c, part of d and f)

c) Skriver om $p(r,t) = P_0 e^{j\omega t}$, $P_0 = B\frac{1}{r}e^{-jkr}$

$v(r,t) = V_0 e^{j\omega t}$, $V_0 = \frac{B}{\rho_0 c}\frac{1}{r}\left[1-\frac{j}{kr}\right]e^{-jkr}$

$$Re\{p\} = \frac{1}{2}(p+p^*) = \frac{1}{2}\left(P_0 e^{j\omega t} + P_0^* e^{-j\omega t}\right)$$

$$Re\{v\} = \frac{1}{2}(v+v^*) = \frac{1}{2}\left(V_0 e^{j\omega t} + V_0^* e^{-j\omega t}\right)$$

$$Re\{p\}\cdot Re\{v\} = \frac{1}{4}\left(P_0 V_0 e^{2j\omega t} + P_0 V_0^* + P_0^* V_0 + P_0^* V_0^* e^{-2j\omega t}\right)$$

Integrating this over 1 period we get:

$$\overline{I(r)} = \frac{1}{4}\left(P_0 V_0^* + P_0^* V_0\right) = \frac{1}{4}\cdot 2\cdot Re\{p\cdot v\} = \underline{\frac{1}{2} Re\{p(r,t)v(r,t)\}}$$

Using this we can calculate $\overline{I(r)}$:

$$\overline{I(r)} = \frac{1}{2}\frac{|B|^2}{\rho_0 c}\frac{1}{r^2} \Rightarrow |B| = B\cdot B^* = \underline{\frac{|V(a)|^2\rho_0^2 c^2 k^2 a^4}{1+ka^2}}$$

$$\Rightarrow \overline{I(r)} = \frac{1}{2}\frac{|V(a)|^2\rho_0 c k^2 a^4}{(1+ka^2)r^2} = \underline{\frac{1}{2}\left(\frac{ka}{kr}\right)^2\frac{|V(a)|^2\rho_0 c k^2 a^2}{1+ka^2}}$$

Vil si kraften en lyd bølge bærer med seg per område i retningen vinkelrett på området. Blir også gjerne kalt lyd energy flukks tetthet.

d) $P_a = \int_S \overline{I(r)}\,dS = \overline{I(r)}\cdot 4\pi r^2 = 2\pi a^2 \frac{|V(a)|^2\rho_0 c(ka)^2}{1+(ka)^2}$

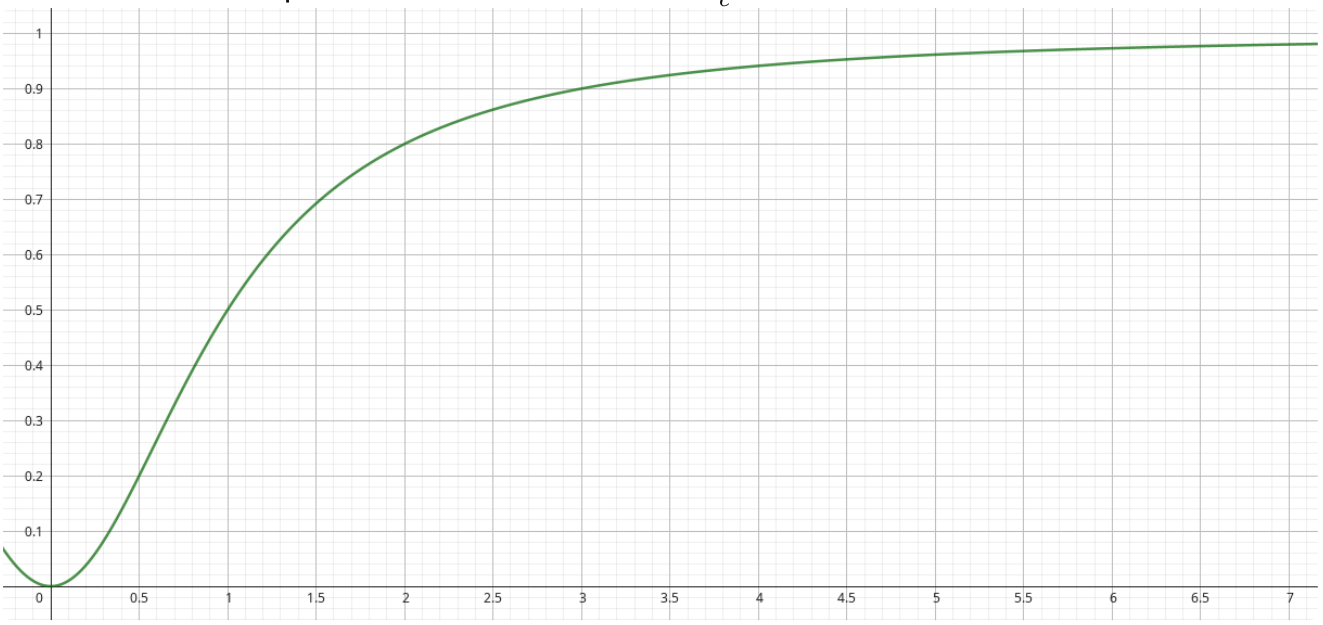
$P_{max} = 2\pi a^2 |V(a)|^2 \rho_0 c \Rightarrow \frac{P_a}{P_{max}} = \frac{(ka)^2}{1+(ka)^2}$ | Plotter normalisert effekten med (ka), $k=\frac{1}{a}$

f) $s/a = 1/100 \Rightarrow s = \frac{a}{100}$ $|V(a)| = w\cdot s \Rightarrow w_0 \cdot s = \frac{c}{a}\cdot\frac{a}{100} = \frac{c}{100}$

$$P_a = 2\pi\, 0{,}1^2 \left|\frac{343}{100}\right|^2 \cdot 1{,}21\cdot 343\cdot\frac{1}{2} \approx \underline{153{,}4\,W}$$

d)

Plot of normalized power as a function of $ka = \frac{\omega a}{c}$:

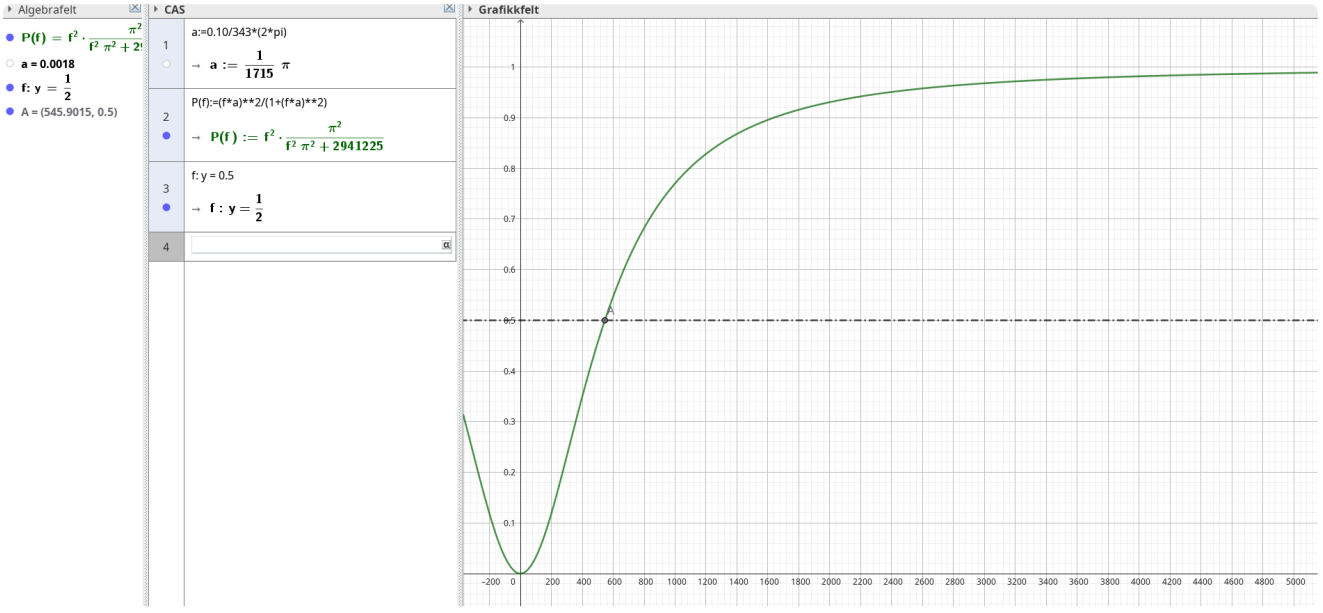


We can see that for low frequencies, the total radiated power is very low, and that as the frequencies increase it flattens out at 100%.

e)

Using the same normalized graph but with:

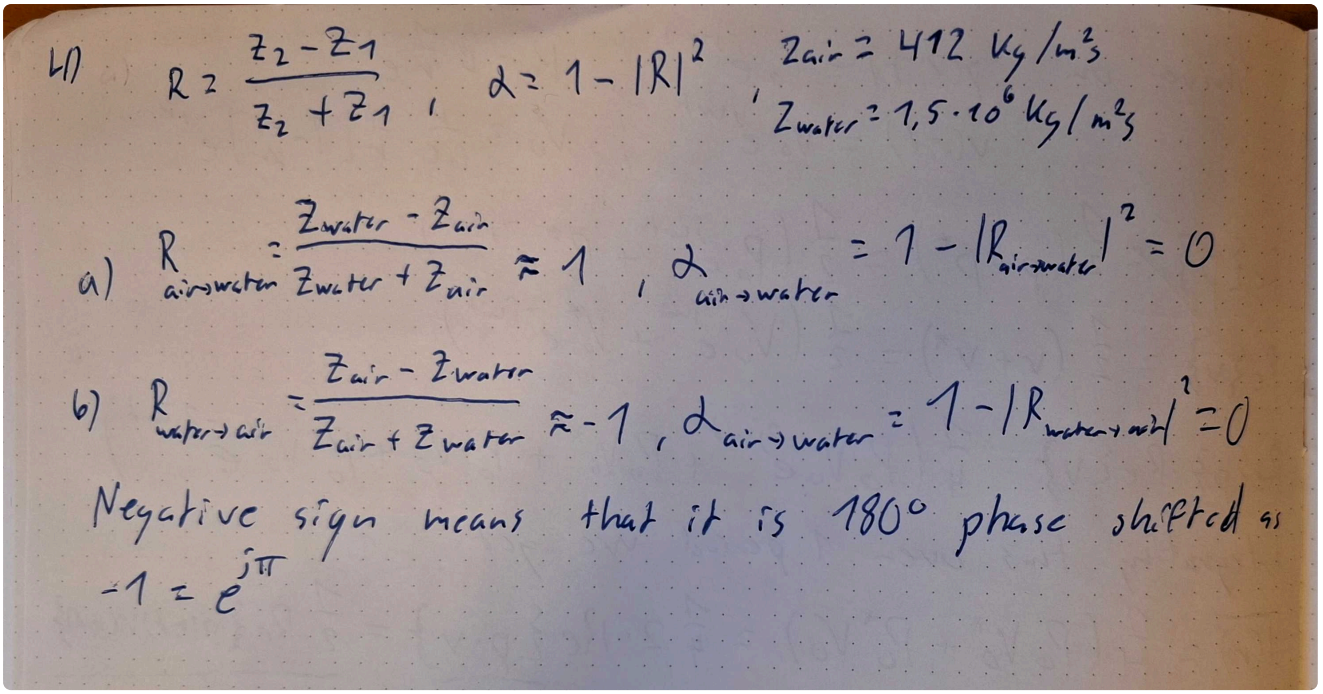
$$ka = \frac{\omega a}{c} = \frac{2\pi f \cdot 0.1}{343}$$



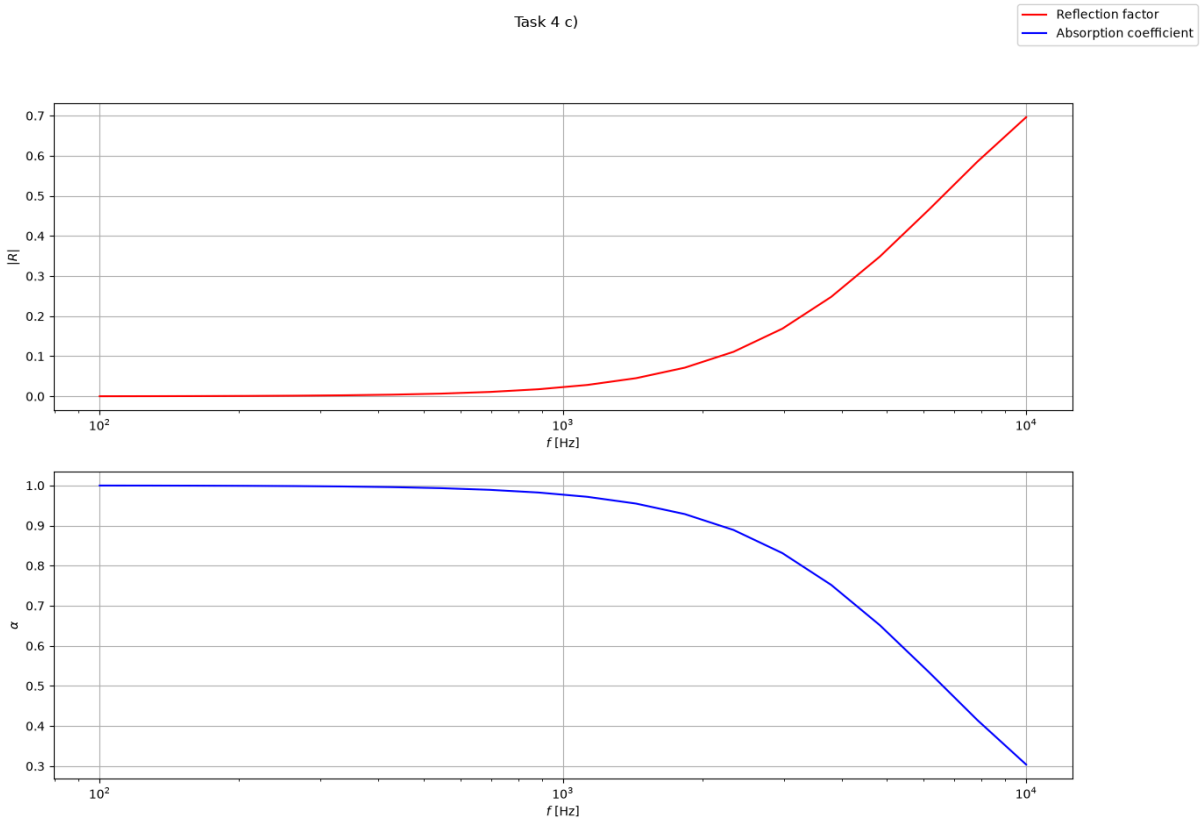
Marking out at 50% we get the frequency 545.9 Hz.

Problem 4)

a and b)



c)



This absorber is affected performance wise by the plastic foil. Deacresing values of m by around $1/4$ improves performance between 10^3 and 10^4 Hz by around 7 times. However that would require making a even thinner plastic foil which could lead to this

simple equation not holding anylonger or other material issues. All in all, still very good noice insulation, as high frequency noise is less common.